AI-PREDICTED – NOT AN OFFICIAL QUESTION PAPER (FOR REVISION & PRACTICE ONLY)

MAHARASHTRA STATE TECHNICAL UNIVERSITY (AUTONOMOUS)

Bachelor of Technology (B.Tech) - Semester IV

SUBJECT: DATABASE MANAGEMENT SYSTEMS & INFORMATION ARCHITECTURE (CS-402)

End Semester Examination - 2025-2026

Time: 3 Hours **Max Marks: 100**

Instructions to Candidates:

1. All main questions carry specified marks. Answer according to section guidelines.
2. Figures to the right indicate full marks.
3. Assume suitable data if necessary and state them clearly.
4. Note: Supplementary questions marked (A), (B), (C) are additional AI-predicted alternative practice variations and do not add to official total marks.

--- SECTION A ---

Q.1 Demonstrate the use of GROUP BY and HAVING clauses in SQL. Differentiate between WHERE and HAVING with queries. **[20]**

Topic: Aggregate Functions & Grouping | Difficulty: Medium

ML Prob: 85.6% (High Probability) | Historical Years: 2012, 2016, 2019, 2022, 2024

SUPPLEMENTARY AI PREDICTED ALTERNATIVES:

Q.1(A) Explain Inner Join, Left Outer Join, Right Outer Join, and Full Outer Join with SQL syntax and sample tables. ***[20]***

Topic: Complex Joins & Subqueries | Diff: Medium

ML Prob: 84.3% (High Probability) | Supporting: 2011, 2014, 2017, 2020, 2023, 2025

Q.1(B) What is a Database Trigger? Explain Before and After triggers with a realistic inventory audit log example. ***[20]***

Topic: Views and Triggers | Diff: Hard

ML Prob: 84.2% (High Probability) | Supporting: 2013, 2017, 2021, 2024

Q.1(C) Explain Clustered vs Non-Clustered Indexes and how the SQL query optimizer generates an execution plan. ***[20]***

Topic: Indexes & Query Optimization | Diff: Hard

ML Prob: 82.0% (High Probability) | Supporting: 2015, 2018, 2022, 2025

Q.2 Discuss ACID properties in transaction processing. Explain how each property is preserved by the DBMS recovery manager. **[20]**

Topic: ACID Properties | Difficulty: Medium

ML Prob: 79.3% (High Probability) | Historical Years: 2011, 2014, 2016, 2019, 2022, 2024

SUPPLEMENTARY AI PREDICTED ALTERNATIVES:

Q.2(A) Explain Entity-Relationship (ER) modeling with an example. Draw an ER diagram for a Hospital Management System. ***[20]***

Topic: ER Modeling & Entities | Diff: Medium

ML Prob: 79.2% (High Probability) | Supporting: 2011, 2013, 2016, 2019, 2022, 2024

Q.2(B) State the definition of BCNF. How does BCNF differ from 3NF? Give a relation that is in 3NF but not in BCNF. ***[20]***

Topic: Boyce-Codd Normal Form (BCNF) | Diff: Hard

ML Prob: 78.9% (High Probability) | Supporting: 2012, 2016, 2019, 2022, 2024

Q.2(C) Explain 1NF, 2NF, and 3NF with suitable anomalies (Insertion, Deletion, Update anomalies). Normalize an unnormalized relation. ***[20]***

Topic: 1NF, 2NF, 3NF Normal Forms | Diff: Hard

ML Prob: 78.4% (High Probability) | Supporting: 2011, 2013, 2015, 2018, 2021, 2023, 2025

--- SECTION B ---

Q.3 Explain the Strict Two-Phase Locking (2PL) protocol. How does 2PL ensure serializability? **[20]**
Topic: Two-Phase Locking (2PL) Protocol | Difficulty: Hard
ML Prob: 78.1% (High Probability) | Historical Years: 2012, 2015, 2018, 2021, 2023, 2025

SUPPLEMENTARY AI PREDICTED ALTERNATIVES:

Q.3(A) Explain the structure of B+ Tree. Demonstrate insertion of keys into a B+ tree of order 4. ***[20]***
Topic: B-Trees & B+ Tree Indexing | Diff: Hard
ML Prob: 77.7% (High Probability) | Supporting: 2012, 2015, 2018, 2021, 2024

Q.3(B) Explain Relational Algebra operations: Select, Project, Cartesian Product, and Natural Join with formal mathematical notation. ***[20]***
Topic: Relational Algebra | Diff: Hard
ML Prob: 76.8% (High Probability) | Supporting: 2014, 2017, 2020, 2023, 2025

Q.3(C) Define Primary Key, Foreign Key, and Candidate Key. Explain Referential Integrity with examples. ***[20]***
Topic: Relational Integrity Constraints | Diff: Easy
ML Prob: 76.4% (High Probability) | Supporting: 2012, 2015, 2018, 2021, 2023

Q.4 Describe Deadlock handling mechanisms: Wait-Die scheme, Wound-Wait scheme, and Wait-For Graph (WFG) detection. **[20]**
Topic: Deadlock Detection & Prevention | Difficulty: Hard
ML Prob: 76.3% (High Probability) | Historical Years: 2013, 2017, 2020, 2024

SUPPLEMENTARY AI PREDICTED ALTERNATIVES:

Q.4(A) Explain Lossless Join Decomposition and Dependency Preservation with a step-by-step testing algorithm. ***[20]***
Topic: Lossless Join Decomposition | Diff: Hard
ML Prob: 75.1% (High Probability) | Supporting: 2014, 2017, 2020, 2023

Q.4(B) Explain Static vs Dynamic Extensible Hashing with directory expansion and bucket splitting. ***[20]***
Topic: Hashing Techniques | Diff: Medium
ML Prob: 75.1% (High Probability) | Supporting: 2014, 2017, 2020, 2023

Q.4(C) State and explain the CAP Theorem. Compare Document-oriented (MongoDB) vs Key-Value (Redis) datastores. ***[20]***
Topic: CAP Theorem & NoSQL Datastores | Diff: Medium
ML Prob: 74.7% (Moderate Probability) | Supporting: 2016, 2019, 2022, 2025

Q.5 Discuss mapping cardinalities (1:1, 1:N, M:N) and structural constraints in ER diagrams. **[20]**
Topic: Cardinality & Keys | Difficulty: Medium
ML Prob: 70.6% (Moderate Probability) | Historical Years: 2013, 2018, 2024